

system receives might vary, due to the use of different quality microphones.

But, despite the limitations, voice recognition does have a lot going for it. The biggest is the cost saving as you do not need any special equipment. Any regular microphone will suffice as an input source. The added benefit is that voice recognition is the only technology that can give remote access to users and hence can have wide ranging applications from phone banking to remote login to secure servers but for that, the issues mentioned above need to be ironed out.

Can Biometrics Be A Threat?

The single biggest argument against Biometrics is that it is a potential threat to privacy. The case is of "Big Brother" breathing down your neck every time you turn around and do even something as mundane as boarding a train to office.

The concern is genuine since individuals would most certainly lose their anonymity in a biometric dependent system. Privacy advocates claim that citizens have a right to anonymity



A US Soldier performs an iris scan on an Iraqi

while engaging in transactions such as online purchasing or in-person voting. If biometrics is used for authentication and identification (the line between these is being fast blurred by biometrics), our anonymity when using different systems or services is lost.

Biometrics would ensure accurate cross-database matching and extensive records about a person's habits, preferences and frequently visited places. As is the case today, this information database can then be sold to second and third parties, making sensitive or private information blatantly public and leading to invasive marketing campaigns and widespread harassment.

Even our movements from office to work would not remain private and could be tracked easily using various security cameras deployed on the road. A case in point is the cameras deployed on certain traffic signals in New Delhi that click a picture of your vehicle when you commit an offence such as jumping a traffic light. The police then send the picture along with a *challan* (traffic ticket) mentioning the date and time for you to pay the fine.

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This could be extended to recognising your face and knowing your background. Judging by the rampant and all-pervasive corruption that we have, is it advisable to give such personal information to someone who can and probably will use it in any which way they like? Maybe not.

Privacy extremists extend this to a time when the computers matching the faces and giving backgrounds would themselves be smart enough to "know" who and what we are and harass us. This doomsday conspiracy, although interesting, is not likely to happen for sometime to come, but should certainly be thought of when implementing biometric solutions.

Biometrics For You

Biometrics is no longer the stuff sci-fi dreams are made of but is here in reality and lots of products incorporate one or another form of biometric security.

Microsoft recently launched a fingerprint scanner, which is compatible with Windows XP. This allows you to link your XP account with your fingerprint. So, instead of entering your password (at the XP logon or at any password websites) you just need to let it scan your fingerprints.

Even laptops from manufacturers like Fujitsu (S7010) and the Lenovo (previously IBM) T42 and the X-series tablet PC (costing over Rs 1,00,000) incorporate built in fingerprint recognition systems and replace the need for you to enter and remember any passwords!

India has not been left behind by the Biometric wave and we do have a few companies dedicated to Biometrics-based product development. Pune-based BioEnable (www.bioenable.co.in) and Mumbai-based Jaypeetex (www.jaypeetex.com) have products ranging from door locks to attendance registers to car immobilisers to computer mice. The Tirupati temple in Andhra Pradesh has deployed biometrics for crowd control and is also looking into incorporating it as a security measure.

During the recently concluded elections in Iraq, a system known as the Biometrics Automated Toolset (BAT) was extensively used to profile the local population. The BAT system stores a person's biometric data such as retina scans, fingerprints, facial data, and links it to names. This was incredibly helpful in, firstly, ensuring security and integrity of registration and the individual and secondly, in avoiding duplication as even with the help of an interpreter, names can be differently or incorrectly spelt.

A pilot project is on in the US where instead of swiping your credit card, you simply put your thumb in the fingerprint scanner in the ATM. This gets matched to the thumbprint in their database and the transaction is processed. Simple, secure and hassle-free. Just what technology should be.

The concerns over biometrics are not without reason. Biometrics can certainly be a powerful security tool to combat terrorism. In the end it is upto the people to decide whether the price of losing their anonymity is justified in order to gain the comfort of security. ■

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